

PAPER CODE	EXAMINER	DEPARTMENT	TEL
INT202			

2nd SEMESTER 2021/22 RESIT EXAMINATION

UNDERGRADUATE – YEAR 3

COMPLEXITY OF ALGORITHMS

TIME ALLOWED: 2 HOURS

INSTRUCTIONS TO CANDIDATES

- 1. This is an open-book examination.**
- 2. The examination will be conducted in all other respects in accordance with the Regulations for the Conduct of Examinations.**
- 3. Total marks available are 100. This exam will count for 100% in the final assessment.**
- 4. Answer all questions. There is NO penalty for providing a wrong answer.**
- 5. Only English solutions are accepted.**

Question 1. [6+6=12 MARKS]

(1) Suppose we are given a sequence S of n elements, each of which is an integer in the range $[0, n^2]$. Describe a simple method for sorting S in $O(n)$ time (no need to write pseudo-code).

[6 marks]

(2) Outline a linear time algorithm to find the strongly connected components of a directed graph (no need to write pseudo-code).

[6 marks]

Question 2. [10+10=20 MARKS]

A company Director General (DG) wishes to organize a party in order to celebrate an award she received. In her company, the hierarchical relationships form a tree in which the DG is the root. The DG wishes that an employee is not invited at the same time with his or her direct "boss". Furthermore, each employee is associated with a score depending on his or her "likability".

a. How to form the list of participants so as to maximize the sum of likability scores?

[10 marks]

b. Given the following sample input, calculate the maximum likability score.

[10 marks]

As for the sample input,

the first line n is the number of the employees;

the second line lists the likability of each employee to the party;

and in the remaining n lines,

the first number is the employee ID,

the second number is the number of subordinates,

and the remaining numbers are the subordinates' IDs.

The sample Input is:

```
6
20 3 5 4 5 30
0 2 2 3
2 1 1
3 1 4
1 0
4 1 5
5 0
```

Question 3. [3+7+7+3=20 MARKS]

Use the following integer elements:

9, 2, 5, 6, 0, 7, 1, 4, 3, 8

to solve the following problems.

(1) Draw the tree representation of the binary min-heap that results when all of the above elements are added (in the given order) to an initially empty heap. Circle the final tree that results from performing the additions. Also show the final array representation of the heap array.

[3 marks]

(2) Given your heap from part a, remove the root from the heap and draw its state afterward. Circle the tree that results after the root is removed. What is the running time (in Big-Oh notation) of a removal operation? Please justify your answer.

[7 marks]

(3) Given your heap from part a, insert -2 into the heap and draw its state afterward. Circle the tree that results after the -2 is inserted. What is the running time (in Big-Oh notation) of an insertion operation? Please justify your answer.

[7 marks]

(4) What is the running time (in Big-Oh notation) of a heap-sort algorithm? Please justify your answer.

[3 marks]

Question 4. [8 MARKS]

Consider the Knapsack problem where we are given a set S of n items such that each item i has a positive benefit b_i and a positive weight w_i , and we wish to find the maximum-benefit subset that does not exceed a given weight W . Show that the greedy algorithm does not always find an optimal solution of the Knapsack problem.

[8 marks]

Question 5. [8+7+5=20 MARKS]

a. For the pair of integers (a,b) where $a = 90$ and $b = 56$, find $d = \gcd(a,b)$ and find a pair of numbers u and v such that $d = au + bv$.

[8 marks]

b. The Caesar army was composed of more than 2000 soldiers and less than 3000. When counted in groups of 11, the remainder was 1; in groups of 9, the remainder was 5, and in groups of 13, the remainder was 8. How many soldiers were there in that army? Explain your

reasoning.
[7 marks]

c. In the RSA method, suppose that $p = 5$, $q = 17$, and $e = 13$. Find the private key d corresponding to these parameters.
[5 marks]

Question 6. [5+15=20 MARKS]

Let Φ be a set of m clauses over the Boolean variables $x_1, x_2, \dots, x_n$ such that each clause contains at most two literals. For a given integer k , the MAX2SAT problem consists in deciding whether there exists a truth assignment satisfying at least k clauses.

(1) Show that MAX2SAT is in NP.
[5 marks]

(2) Derive a reduction from 3SAT to MAX2SAT. Is MAX2SAT is NP-Complete? Please justify your answer.
[15 marks]

END OF EXAM PAPER